

Water: Spring, Cistern, Treatment

Carpathian OER Commons — Utilities Module

Michel Garand | 2026-09-27 | Version v0.2 | CC BY-SA 4.0

Maturity: L1 — designed and costed, untested. Every cost line is an estimate, labelled as such in Section 6. Read every figure below at this level and no higher.

Published under Section 11 of the pathway paper. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only.

Converted from the water working paper, EN v0.1, Sections 1 to 5, 8, and 9. Its rainwater section becomes the rainwater and nursery irrigation module. Its greywater section is superseded by the sanitation working paper, EN v0.2, and belongs to the greywater, reed bed, and pond module.

v0.2: capacity revised from 25 to 34 people, 2026-09-27. The source working paper is sized at 25; every quantity below that depends on group size has been recalculated at 34.

1. Purpose

Drinking, cooking, and washing water for everyone on site at capacity, in every season and through power outages, with a reserve in the shelter. Spring, gravity, sand, and stone — built by local trades and repairable by the people who use it.

Four rules. Gravity before pumps: no step that supplies people depends on electricity. Two waters kept apart: spring water for people, roof water for plants, sharing no pipe. Storage covers the gap: three days of demand held, so a dry spell, a frozen line, or a repair does not stop a session. Treatment assumes contamination will happen.

Connections. Depends on: the spring measurement; the enquiries on the protection zone and a fire-fighting reserve; the energy module for the UV unit on the essential circuit. Feeds: the greywater, reed bed, and pond module with all used water; the shelter with its reserve; the rainwater module takes the non-drinking uses when the spring falls between thresholds.

Patterns. Grows from: not yet named. How homesteads here stand to their spring, and whether this spring is held in any way that decides how it may be touched, are questions for the elders before the spring box is built. Feeds: not yet named. Resonates with: 207 Good Materials — stone from the site or the valley, built and repaired by local trades; 83 Master and Apprentices — the mason's work beside toloka labour; 104 Site Repair — the cistern and the catchment fence on the poorer ground. In tension with: 64 Pools and Streams — a buried cistern and a piped main take water out of sight; the spring and its catchment are where it can stay visible. 25 Access to Water — the catchment fence keeps people back from water the pattern would open to them.

2. Context of validity

Condition	Range this design holds for
Climate	[set] — mountain; frost depth assumed under 1 m
Altitude	[set]
Group size	capacity: 34 people, the example below
Season of use	all seasons, including frozen ground
Water source	a spring giving at least 1.8 L/min in the driest month
Sanitation	dry sanitation; flush toilets change the demand figure
Terrain	ground allowing a cistern 10 to 20 m above the highest tap

3. Sizing method

What you must measure yourself before using this module:

- **Spring yield in the driest month** — late August, or a hard February with frozen ground. A bucket of known volume and a stopwatch, three readings, keep the lowest. Record the date and the weather of the preceding weeks.
- **Height difference** between spring, cistern site, and highest tap, and the real pipe lengths — from a site walk.
- **Your group size at capacity** and your sanitation type.

Choose a daily figure per person.

Level	L per person per day
Survival minimum, humanitarian standard	15
Basic health and hygiene, WHO threshold	50
Rustic residential, dry sanitation	60–80
Institutional, flush toilets and showers	100–150
European household average	130–145

The household figure is not a design number for a site with dry sanitation and communal washing. The European Environment Agency puts direct household use at roughly 130 to 145 L per person per day, a quarter to a third of it toilet flushing.

Our example: water_per_person, 75 L a day, everyone on site alike, kitchen included; capacity, 34 people; storage_days, 3. Values from the site parameters file, held in the workshop.

Quantity	Calculation	Our example
Daily demand	people $\times$ L per day	$34 \times 75 = 2,550$ L
Storage, three days	$3 \times$ daily demand	7,650 L, built as 8 m ³
Spring yield, bare minimum	daily demand $\div$ 1,440 min	1.8 L/min
Spring yield, design	minimum $\times$ 2, rounded up	4 L/min
Shelter reserve, 48 h	people $\times$ 3 L $\times$ 2 days	204 L
Sand filter area	daily volume $\div$ rate	0.5–1.1 m ²
Pressure at the tap	about 1 bar per 10 m of height	1–2 bar

The sand filter figure takes 2.55 m³ a day at a filtration rate of 0.1 to 0.2 m per hour. The 1 m² costed in Section 6 is enough at the faster rate and short at the slower one: 1.1 m² is needed to filter at 0.1 m per hour. The 8 m³ cistern holds 3.1 days at 34 people, so its size and cost stand.

Read the spring against the demand.

- 4 L/min or more: the spring carries the site alone.
- Between 1.8 and 4 L/min: it carries the people only if rainwater takes every non-drinking use it can.
- Below 1.8 L/min: a second source is needed before anything is built.

4. Bill of materials

Quantities for the 8 m³ example. Substitutions are alternatives named in the design, not yet known to work here.

Item	Quantity	Alternatives in the design
Cement, 25 kg bags	92	none named
Sand	about 5.5 m ³	none named

Item	Quantity	Alternatives in the design
Crushed stone and gravel bed	about 4 m ³	none named
Rubble stone for walls	about 13 m ³	none named
Rebar, two slabs	about 250 kg	stone barrel vault, no slab
Crystalline waterproofing	potable-rated	none named
Manhole covers, lockable	2, 60 × 60 cm	none named
PE pipe 32 mm PN10	about 250 m, assumed	none named
Slow sand filter media	graded sand, gravel	none named
UV unit	1, drinking and kitchen	none named
Food-grade containers	204 L in all	none named

The cistern, as designed.

Element	Specification
Internal footprint	3.4 × 2.0 m, divided by a 0.3 m wall
Each chamber	1.55 × 2.0 m
Water depth	1.3 m
Freeboard	0.3 m, internal height 1.6 m
Outer walls	rubble stone in cement mortar, 45 cm
Floor	reinforced concrete, 15 cm, on 10 cm gravel
Roof	reinforced concrete slab, or a stone barrel vault
Earth cover	at least 1 m, below the frost line
Lining	three coats of cement render, about 2.5 cm in all

5. Build

The spring box, supply main, sand filter, and potable cistern are one system and are built first.

Step	Trade	Time	If done out of order
Measure the spring	household	dry month	everything is mis-sized
Spring box, about 1 m ³	mason	[from L2]	[from L2]
Catchment fence	household	[from L2]	[from L2]
Trench and supply main	excavator	[from L2]	[from L2]
Slow sand filter	mason	[from L2]	[from L2]
Cistern excavation	excavator	about 10 h	[from L2]
Cistern walls and slabs	mason	[from L2]	[from L2]
Render and waterproofing	mason	[from L2]	[from L2]
Backfill and earth cover	excavator	about 4 h	[from L2]
Fill, flush, refill 2–3×	household	[from L2]	lime in the water
Laboratory test	laboratory	[from L2]	no record for inspection

Details that decide whether it works. The spring box: a collection chamber, a screened overflow, a lockable lid, a clean-out, the ground around it sealed with clay and sloped so rain runs away from the box. The cistern: inlet through a settling box; outlet 10 to 15 cm above the floor; a washout at the lowest point; an overflow with a flap valve; a screened vent; one lockable manhole per chamber; walls meeting the floor in a rounded cove so the render does not crack at the corners.

Structure. A buried cistern carries earth pressure when empty and water pressure when full. A local builder or engineer confirms the walls against the soil before building.

6. Cost

Estimates, not quotations. Month: autumn 2026. Place: not a quotation; local anchor prices. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

Anchors: cement about 215 UAH for a 25 kg bag delivered; rubble masonry labour 3,000 UAH per m³, inside a range of about 1,500 to 3,500 UAH quoted before 2025 and adjusted upward; excavator hire 1,800 UAH an hour.

6.1 Spring capture

Item	UAH	EUR	Basis
Spring box, stone and cement, about 1 m ³	14,000	269	estimated
Screened overflow and clean-out	3,000	58	estimated
Fence around the catchment, about 80 m	20,000	385	estimated
Subtotal	37,000	712	

6.2 Supply main

Item	UAH	EUR	Basis
PE pipe 32 mm PN10, about 250 m	11,250	216	length assumed
Valves, fittings, air release	6,000	115	estimated
Trenching 1 m deep, about 250 m	30,000	577	length assumed
Subtotal	47,250	909	

6.3 Potable cistern, 8 m³

Item	UAH	EUR	Basis
Cement, 92 bags of 25 kg	19,780	380	estimated
Sand, about 5.5 m ³	6,050	116	estimated
Crushed stone and gravel bed, about 4 m ³	5,200	100	estimated
Rebar for two slabs, about 250 kg	10,000	192	estimated
Formwork or vault centering	6,000	115	estimated
Crystalline waterproofing, potable-rated	10,000	192	estimated
Two lockable manhole covers	7,000	135	estimated
Inlet, outlet, washout, overflow, vent	8,000	154	estimated
Excavator, about 10 hours	18,000	346	estimated
Mason: stone walls, about 10 m ³	30,000	577	estimated

Item	UAH	EUR	Basis
Slabs: concrete, rebar, formwork labour	12,000	231	estimated
Internal render, about 26 m ²	10,400	200	estimated
Backfill and earth cover, about 4 hours	7,200	138	estimated
Subtotal	149,630	2,878	

6.4 Treatment

Item	UAH	EUR	Basis
Slow sand filter, 1 m ² , stone box	12,000	231	at 25; [set]
Filter sand and gravel, graded	4,000	77	estimated
UV unit, drinking and kitchen line	15,000	288	estimated
Subtotal	31,000	596	

6.5 Shelter reserve and testing

Item	UAH	EUR	Basis
150 L in food-grade containers	1,500	29	at 25; [set]
First laboratory water test	4,000	77	estimated
Subtotal	5,500	106	

6.6 Summary

Section	UAH	EUR
Spring capture	37,000	712
Supply main	47,250	909
Potable cistern	149,630	2,878
Treatment	31,000	596
Shelter reserve and testing	5,500	106

Section	UAH	EUR
Construction subtotal	270,380	5,200
Design and engineer's check, 8 percent	21,600	415
Contingency, 20 percent	58,400	1,123
Total external cost	350,380	6,738

The summary is recalculated from the source lines at the source's own percentages and rounding, because rainwater and greywater have left this module. The source's combined total was 612,080 UAH. Two lines are still costed at 25 people: the sand filter at 1 m² against 1.1 m² needed, and the shelter reserve at 150 L against 204 L. Both are marked for a new estimate; the total does not yet include the difference.

Contribution, not in the total. For the cistern alone: rubble stone from the site or the valley, about 13 m³, valued at 15,600 UAH; carrying, mixing, and general labour by toloka, valued at 20,000 UAH — together 35,600 UAH, about 685 EUR. Other stone structures add stone and labour in proportion; the household sets the figures and the local rate.

Running cost. Laboratory test once a year, about 4,000 UAH. UV lamp replaced yearly, about 1,500 UAH. Sand filter maintenance by the household.

7. Operation and maintenance

Responsible: [from L2 — one named role and a deputy].

- **Daily:** [from L2]
- **Seasonal:** scrape the top centimetres of the sand filter when the flow slows, a few times a year.
- **Annual:** full laboratory water test, kept on file; UV lamp replaced.
- **Cistern cleaning:** one chamber emptied and cleaned while the other supplies. Interval [from L2].

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Contamination between spring and tap	[from L2]	[from L2]

Failure	Early sign	Repair and cost
Spring below 1.8 L/min, dry month	[from L2]	second source
Lime leaching from new cement	[from L2]	flush 2–3×, [from L2]
Sand filter clogging	flow slows	scrape top layer, labour
Render cracking at wall-floor corners	[from L2]	[from L2]
Frozen supply line	[from L2]	[from L2]

The design names the contamination routes: animals in the catchment, a cracked pipe, a lid left open.

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	not checked	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, unchecked. The source states that Ukrainian rules set sanitary protection zones around drinking-water sources, typically 30 to 50 m for the first belt around groundwater, without naming the instrument. The zone for a given source is confirmed with the district sanitary service before the fence line is set. Whether the State Emergency Service requires a separate fire-fighting water reserve is open, and could exceed this system. A children's establishment will be asked for water test results by the sanitary service.

11. Adaptation notes

- **A drier valley:** measure first; below 4 L/min, move every non-drinking use to rainwater; below 1.8 L/min, a second source before anything is built.
 - **A colder valley:** [set] — the design already assumes at least 1 m of earth cover, below the frost line.
 - **A smaller group:** recalculate with Section 3; storage stays at three days of demand.
 - **A larger group:** recalculate with Section 3; the sand filter scales with daily volume.
 - **Flush toilets:** use the institutional figure, 100 to 150 L per person per day.
-

12. Provenance and licence

- **Written by:** Michel Garand. [set — others as they choose to be named]
 - **From whose knowledge:** European Environment Agency household use figures; the WHO basic health threshold; the humanitarian survival minimum; slow sand filtration practice.
 - **Traditional knowledge:** none.
 - **Text:** CC BY-SA 4.0
 - **Drawings and designs:** [hardware licence — open decision]
 - **Measurement data:** none yet
 - **Gate 1:** household decision, 2026-09-27 — module may be published as it stands
 - **Wartime review:** 2026-09-27, Michel Garand — read against pathway Section 11; nothing places the site
-

13. What to check

- **Spring yield** in its driest month, measured as in Section 3.
- **Sanitary protection zone** for the source, from the district sanitary service; name the instrument.
- **Fire-fighting reserve:** whether the State Emergency Service requires one.
- **Height and pipe lengths** from a site walk; the 250 m is assumed.
- **Soil at the cistern site,** checked by a local builder or engineer.
- **Quotations** for the lines in Section 6, dated, to replace the estimates.
- **Climate and altitude** ranges for Section 2.
- **Colder-valley adaptation** for Section 11.
- **From L2, once built:** build times and out-of-order consequences in Section 5; the responsible role, daily tasks, and cleaning interval in Section 7; early signs and repair costs in Section 8; Section 9.

- **Sand filter and shelter reserve** re-estimated at 34 people: 1.1 m² of filter, 204 L of reserve.
 - **The capacity figure**, 34 people: decided 2026-09-27 that it may be published, as a design load and never beside a date.
-

© 2026 Michel Garand | Carpathian OER Commons | CC BY-SA 4.0 stewardship@ubec.network